

Battery Charging and Discharge in CAISO's Spring Duck Curve: A Year-over-Year Comparison

Bryan Lu, IEEE PES Student Member

This whitepaper uses public CAISO data to verify a known trend in the CAISO duck curve: battery storage is increasingly charging during solar-heavy midday hours and discharging during the evening ramp. The analysis compares April and May operating days from 2021 through 2025 and focuses on three measures: average midday charging, average evening discharge, and battery discharge relative to the 5-9 PM calculated net load ramp, where calculated net load means CAISO load minus solar and wind generation. The results show that batteries have become more visible in the daily spring duck-curve pattern, but the analysis is descriptive rather than causal. It does not estimate what CAISO operations would have looked like without battery storage.

1. Introduction

The CAISO duck curve began as a warning about solar generation. High midday solar generation can reduce the load other resources must serve, while falling evening solar generation creates a steep ramp-up in the load other resources must serve. The U.S. Department of Energy describes the duck curve as the difference between electricity demand and available solar energy over the course of the day, with solar abundant during sunny hours and then falling as demand peaks in the evening [1]. CAISO materials define net demand as demand minus wind and solar [2].

The modern CAISO operating pattern now includes a much larger battery fleet. CAISO's Department of Market Monitoring reported that CAISO balancing-area battery storage grew from about 500 MW in 2020 to about 13,000 MW in December 2024, and that batteries provided about 8.6 percent of CAISO balancing-area energy during hours 17-21 in 2024 [3]. EIA describes the same operating logic: batteries can charge with excess midday solar and discharge later as the sun sets [4].

This paper tracks that shift in a narrow way. It does not argue that batteries solved the duck curve or caused the ramp to shrink. It asks whether batteries became more visible in the same spring daily windows that define CAISO's duck-curve challenge: the midday dip and the evening ramp.

2. The Duck Curve, Reliability, and Battery Dispatch

For this paper, the important issue is not the duck shape itself. The important issue is that CAISO must continuously balance supply and demand to maintain grid reliability. When net load rises quickly from late afternoon into evening, flexible resources such as natural gas, imports, hydropower, and batteries may need to increase output or otherwise support the system.

Battery charging means batteries are absorbing electricity from the grid. Battery discharge means batteries are putting electricity back into the grid. Batteries do not generate energy as solar, wind, natural gas, or hydropower do; they shift stored electricity from one period to another.

In CAISO, the timing of battery dispatch is tied to solar generation. During solar-heavy midday hours, the calculated net load is often lower and batteries can charge. Later in the day, solar output falls and calculated net load rises, so batteries can discharge into the evening ramp.

3. Data and Method

This paper uses public CAISO data accessed through GridStatus to compare battery behavior on April and May operating days from 2021 through 2025 [5]. April and May are used because spring conditions often

show the duck curve clearly: solar output is high, while moderate weather can keep total demand lower than during summer peaks. This makes the midday dip and evening ramp easier to observe.

The analysis calculates net load as CAISO load minus solar and wind generation. Battery charging and discharge are analyzed separately because battery behavior is not fully visible from the net-load curve alone. CAISO states that public demand and net-demand trend data do not include battery storage while charging [2].

For each year, the analysis averages April and May operating days by clock hour. This creates average spring daily profiles for calculated net load, solar generation, and battery net output. The three main metrics are average midday battery charging, average evening battery discharge, and battery discharge relative to the 5-9 PM calculated net-load ramp.

Battery discharge relative to the 5-9 PM ramp is a scale comparison. For example, if calculated net load rises by 20 GW from 5-9 PM and batteries discharge an average of 5 GW during that window, then average battery discharge is roughly 25 percent of the ramp size. This does not prove that batteries caused the ramp to shrink; it only shows how large the observed discharge is relative to the ramp CAISO must manage.

4. Results

4.1 Calculated net load and solar output

The calculated net-load profiles show the spring duck-curve shape getting deeper over the comparison period. The midday trough is lower in later years, especially 2024 and 2025, while the evening ramp remains large. The solar profile shows the main reason: average spring solar output is substantially higher in the later years of the sample.

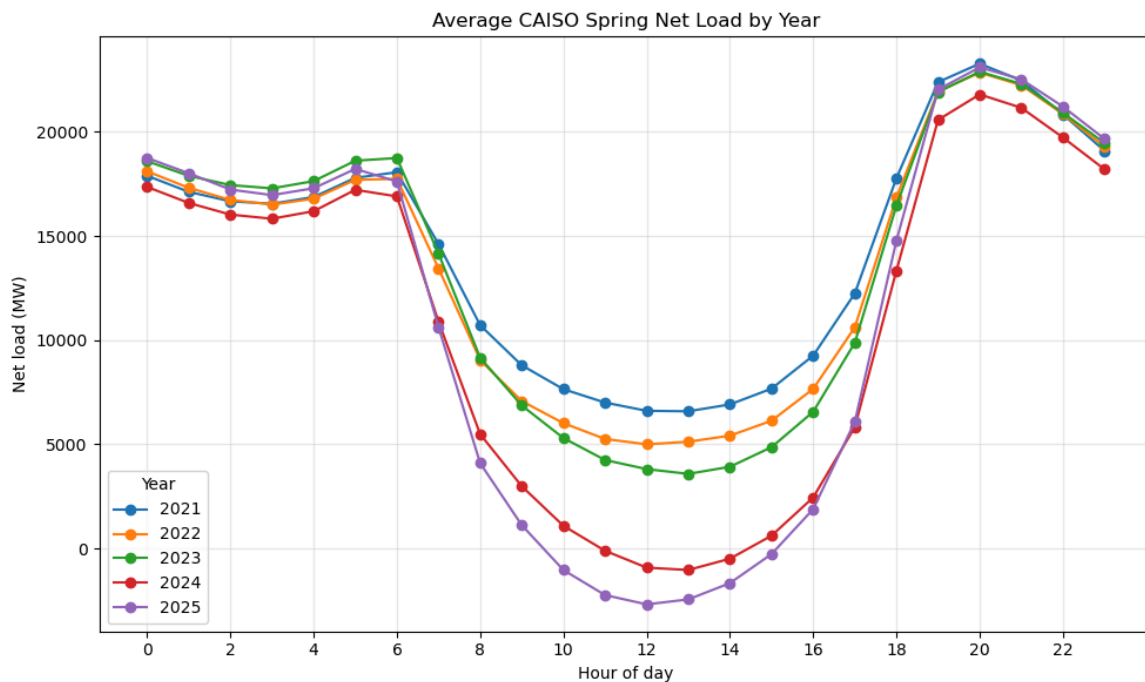


Figure 1. Average CAISO spring calculated net load by hour, 2021-2025. Calculated net load is defined as CAISO load minus solar and wind generation. The chart shows a deeper midday trough in later years and a large evening rise after solar output falls.
Source: Author's analysis of public CAISO data accessed through GridStatus [5]. Values are April/May averages by clock hour for each year.

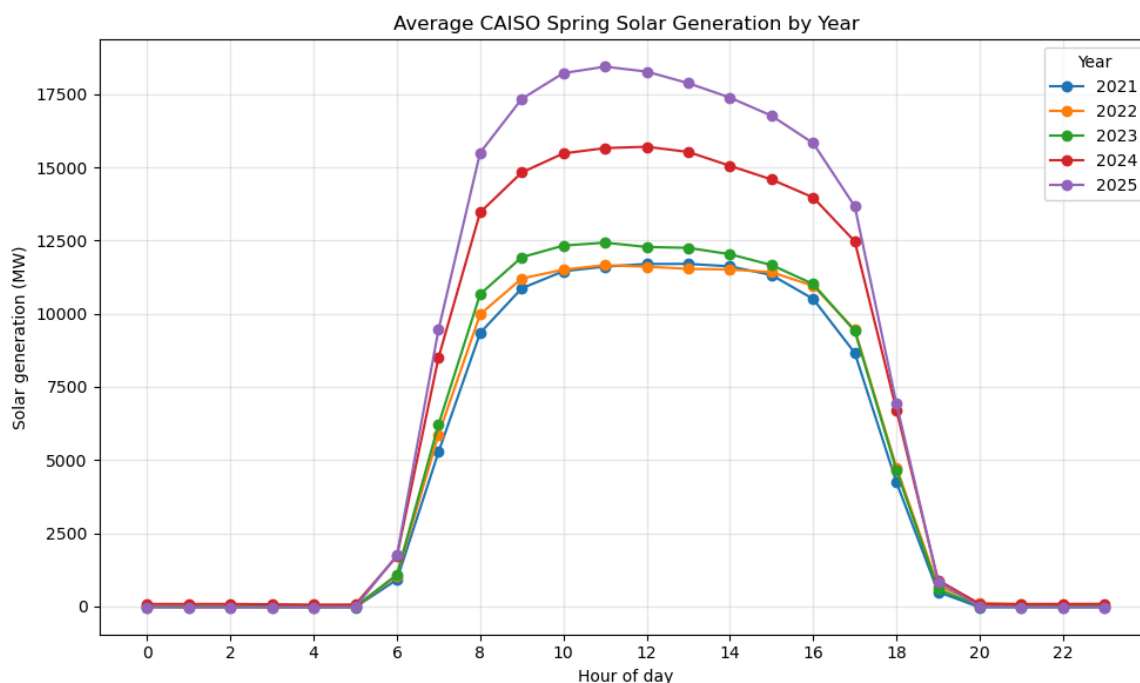


Figure 2. Average CAISO spring solar generation by hour, 2021-2025. Solar output rises substantially in 2024 and 2025, helping explain the lower midday calculated net-load values and the continued importance of the evening transition.

Source: Author's analysis of public CAISO data accessed through GridStatus [5]. Values are April/May averages by clock hour for each year.

4.2 Net battery output

In the net battery output profiles, negative values indicate charging, and positive values indicate discharge. The profiles show a direct trend: batteries increasingly absorb electricity during midday and inject electricity during the evening ramp.

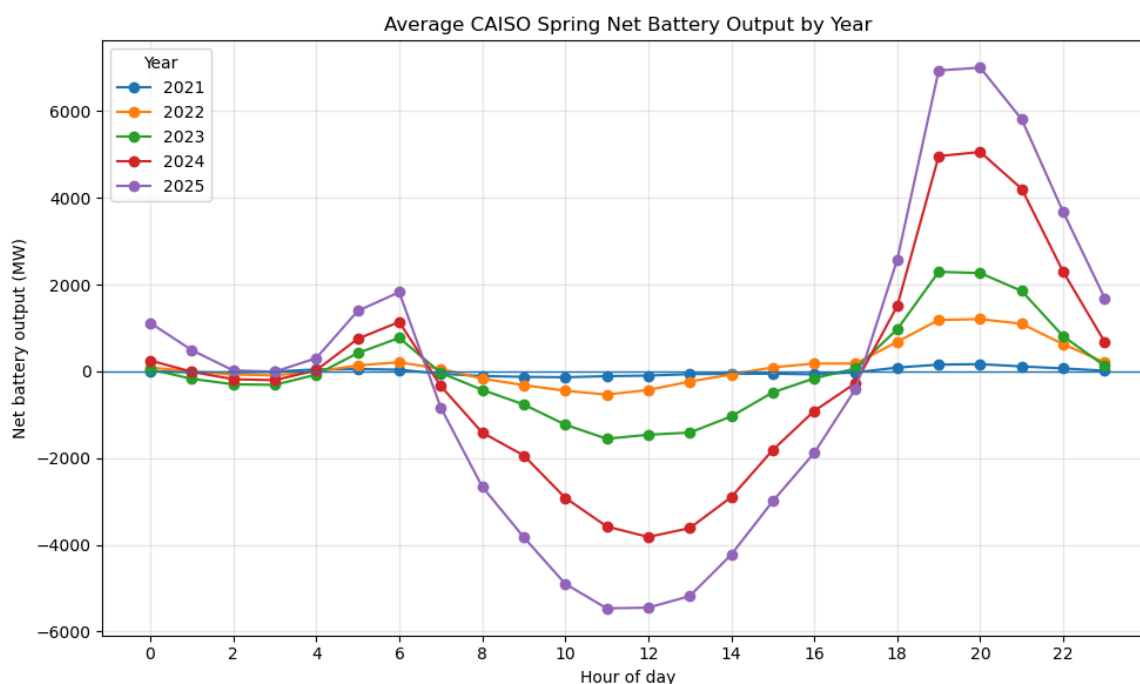


Figure 3. Average CAISO spring net battery output by hour, 2021-2025. Negative values indicate battery charging and positive values indicate battery discharge. The figure shows the storage-shifting pattern directly: midday charging followed by evening discharge.

Source: Author's analysis of public CAISO data accessed through GridStatus [5]. Values are April/May averages by clock hour for each year.

4.3 Midday charging, evening discharge, and ramp scale

The year-over-year bar charts isolate the magnitude of the trend. Average midday battery charging increased from 103.7 MW in 2021 to 5,045.9 MW in 2025. Average evening battery discharge increased from 115.0 MW to 4,188.0 MW over the same period.

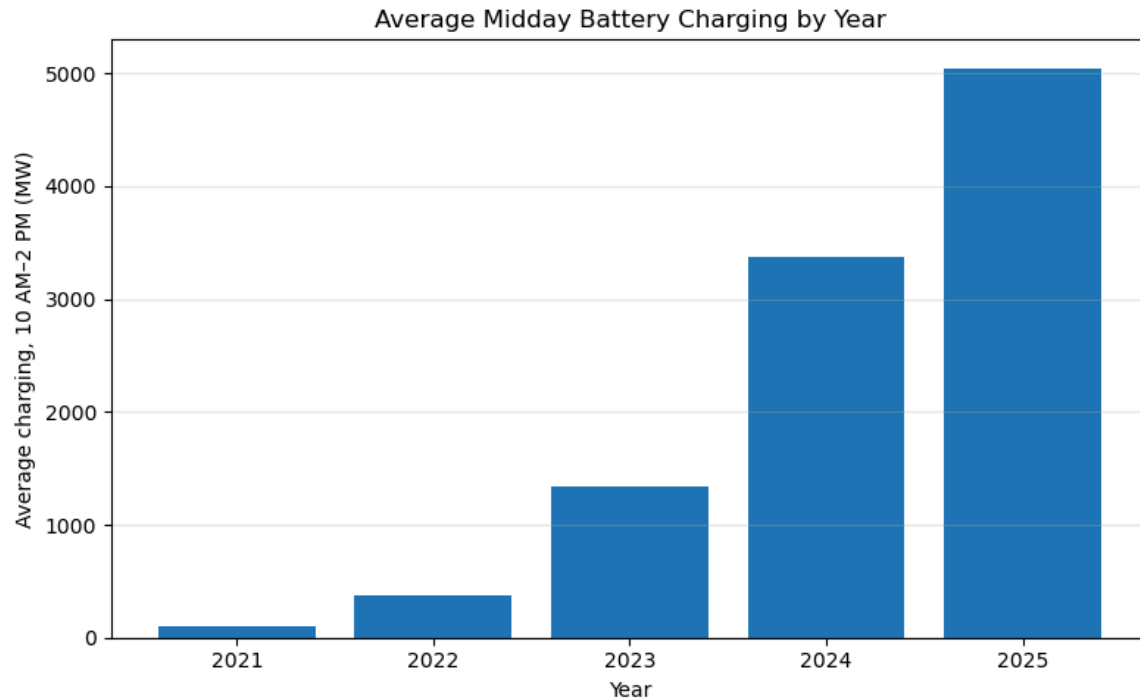


Figure 4. Average midday battery charging by year, 2021-2025. Values are average battery charging during the 10 AM-2 PM window across April and May operating days.

Source: Author's analysis of public CAISO data accessed through GridStatus [5].

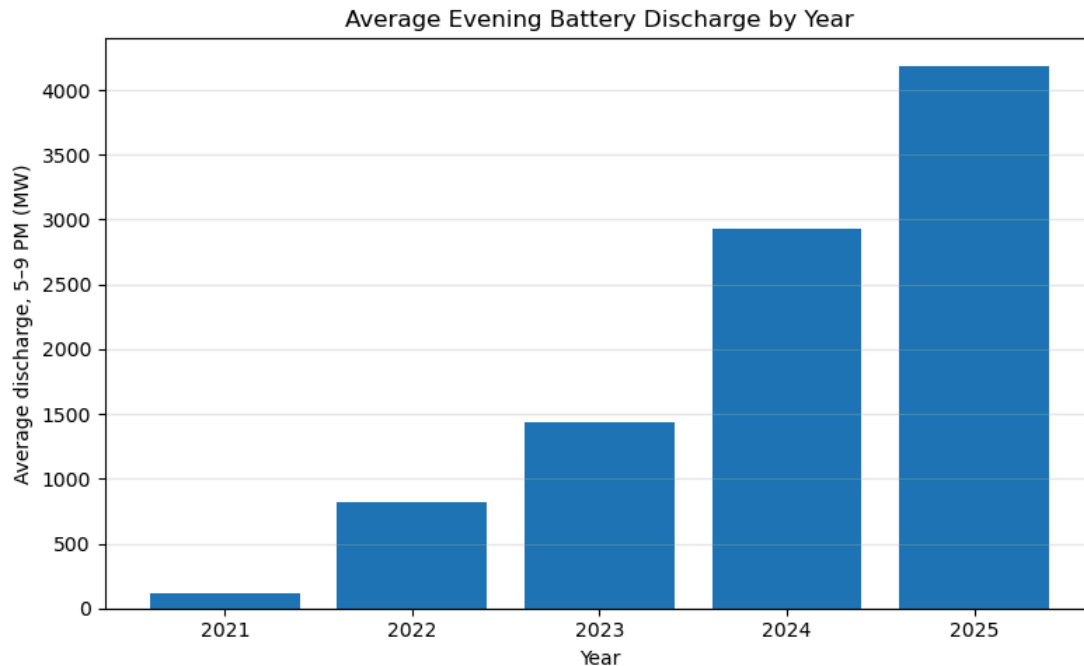


Figure 5. Average evening battery discharge by year, 2021-2025. Values are the average battery discharge during the 5-9 PM window across operating days in April and May.

Source: Author's analysis of public CAISO data accessed through GridStatus [5].

Battery discharge also grew relative to the size of the calculated 5-9 PM net-load ramp. The discharge-to-ramp comparison rose from 1.1 percent in 2021 to 25.5 percent in 2025. This means observed average discharge became large relative to the ramp; it does not mean batteries caused the ramp to shrink.

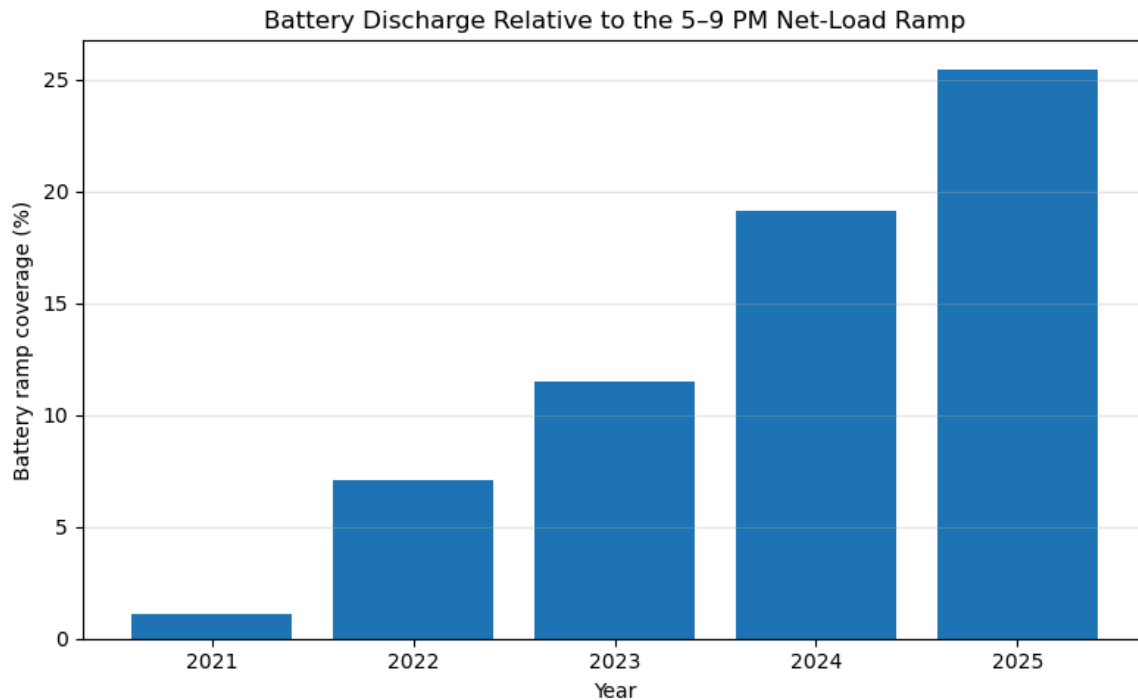


Figure 6. Battery discharge relative to the 5-9 PM calculated net-load ramp, 2021-2025. The metric compares the average 5-9 PM battery discharge with the increase in calculated net load over the same window.

Source: Author's analysis of public CAISO data accessed through GridStatus [5].

Table 1. Key spring battery and calculated net-load metrics, 2021-2025.

Year	Midday charging (MW)	Evening discharge (MW)	5-9 PM net-load ramp (MW)	Discharge relative to ramp (%)
2021	103.7	115.0	10,218.4	1.1
2022	372.6	822.8	11,623.0	7.1
2023	1,342.0	1,431.6	12,422.5	11.5
2024	3,371.5	2,928.1	15,326.4	19.1
2025	5,045.9	4,188.0	16,438.5	25.5

Note: Values are April/May averages by clock hour. Calculated net load is CAISO load minus solar and wind generation. The discharge-to-ramp percentage is a scale comparison, not a causal estimate.

5. Interpretation

The results support a narrow conclusion: batteries are now a visible part of CAISO's spring duck-curve operating pattern. In 2021, average spring battery charging and discharging were small relative to the scale of the net-load curve. By 2025, average midday charging exceeded 5 GW and average evening discharge exceeded 4 GW.

This changes how the modern CAISO duck curve is described. The original problem was solar-driven: high midday solar lowered net load, and falling evening solar created a ramp. That is still true. The difference is that storage is now visibly participating in both sides of the daily pattern. Batteries charge during the midday dip and discharge during the evening ramp.

The results do not show that batteries eliminated the ramp. The 5-9 PM calculated net-load ramp increased from 10.2 GW in 2021 to 16.4 GW in 2025. Still, battery discharge became much larger relative to that ramp, rising from 1.1 percent to 25.5 percent of the ramp scale in this comparison.

6. Limitations

This analysis is descriptive. It does not estimate a counterfactual CAISO system without batteries, nor does it isolate battery effects from changes in weather, load, solar capacity, hydro conditions, imports, generator availability, transmission constraints, or market behavior.

The net-load series is calculated from load, solar, and wind data. Because CAISO states that public demand and net-demand trend data do not include battery storage during charging [2], the paper treats charging and discharging as separate variables rather than assuming they are fully represented in the net-load curve.

The analysis also does not model state of charge, bid-in capacity, awards, ancillary services, or detailed storage market behavior. CAISO publishes daily energy storage reports that include some operational and market metrics, but that level of detail is outside the scope of this first descriptive comparison [6].

7. Conclusion

The CAISO duck curve began as a way to show a solar-driven net-load pattern: a low midday belly and a steep evening ramp. This paper shows why the modern spring duck curve can also be understood as a storage-dispatch pattern layered onto that solar-driven shape.

From 2021 through 2025, average midday battery charging increased from 103.7 MW to 5,045.9 MW, and average evening battery discharge increased from 115.0 MW to 4,188.0 MW. Battery discharge relative to the 5-9 PM calculated net-load ramp increased from 1.1 percent to 25.5 percent. These results verify that batteries have become much more visible during the same hours that define CAISO's duck-curve challenge.

The final takeaway is limited but useful: batteries have not made the duck curve disappear, but they have become a measurable part of how battery storage shifts energy from solar-heavy midday hours toward the evening ramp.

8. References

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[3] California ISO Department of Market Monitoring, "2024 Special Report on Battery Storage," May 29, 2025. [Online]. Available: <https://www.caiso.com/documents/2024-special-report-on-battery-storage-may-29-2025.pdf>. Accessed: Jun. 19, 2026.

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